

Your Workbench, your data

Lab 1: 40 hands-on minutes on v9

September 2026

Everything you touch today is real **Controlled Tier participant data**. Person-level files never leave the workspace. Printed outputs are aggregate, rounded, and suppression-screened by the scripts — keep them that way. There is no answer key: your numbers are real. This handout gives plausibility checks instead.

0–6 minutes: configuration and preflight

```
cd aou_cli_workshop
cp config.aou.example.sh config.aou.sh
nano config.aou.sh          # fill the values projected on screen
export WORKSHOP_CONFIG=config.aou.sh
bash scripts/00_preflight.sh
```

Fill `WORKSHOP_BUCKET`, `BILLING_PROJECT`, and `CDR_DATASET` from the values the instructor projects. `config.aou.sh` is git-ignored — it never gets committed. The `export` line matters in *every new terminal*: it is how the scripts know to run in AoU mode, where the disclosure gate blocks rather than warns.

6–11 minutes: stage genotypes, build the phenotype

```
bash scripts/10_fetch_aou_genotypes.sh
bash scripts/11_build_phenotype.sh
```

While the fetch runs (~1–3 minutes), read the two scripts. Then look at what arrived — two lines each, before trusting anything:

```
head -n 2 lab_data/chr22_filtered.fam    # FID convention?
head -n 2 lab_data/chr22_filtered.bim    # ID scheme?
head -n 2 lab_data/ancestry_preds.tsv    # key column name?
```

Checkpoints: samples in the fileset _____; variants _____.

The phenotype build queries height as program source concept **903133** — not the standard LOINC concept, which would silently pool in EHR heights — for a 1-in-10 teaching subset, then reduces to one row per person (median), applies 100–250 cm bounds, and standardizes within sex. Every one of those clauses is a decision; `scripts/build_aou_pheno.py` states each one.

11–16 minutes: phenotype EDA

```
bash scripts/12_pheno_eda.sh
cat results/aou_pheno_summary.txt
```

Open `results/aou_height_hist.png` and `aou_height_kde_sex.png` from the JupyterLab file browser. **Plausibility checks** (not answers): US-cohort sex-specific height means sit near 176 cm (male) and 162 cm (female) with SDs of roughly 7–8 cm; your subset should be in that neighborhood. Funnel counts print rounded to the nearest 100 — explain to your partner why.

16–24 minutes: score the PGI

```
bash scripts/13_score_real_pgi.sh
cat results/aou_pgi_summary.txt
```

Checkpoints: weight rows matched (variants processed) _____; skipped _____. A very low match rate means a wrong ID scheme or wrong column indices — stop and ask. Open `results/aou_pgi_kde_ancestry`.

This one-chromosome PGI is deliberately partial. What matters is that you counted every match and every join on real data.

24–29 minutes: the regression

```
bash scripts/14_pgi_regression.sh
cat results/aou_pgi_regression.txt
```

The model: standardized height on the standardized PGI, adjusting for age, sex, and five genetic PCs (parsed from AoU’s ancestry file — note its stringified-list column, handled in the script). **Checkpoints:** the PGI coefficient _____; its incremental R^2 _____. Read the headline row aloud in words: “one SD higher in this PGI is associated with ... SD of height, conditional on the covariates.” Then read the three honest labels at the bottom of the output.

29–34 minutes: interpret with a partner

1. Three joins, three key spellings — name them.
2. Why the median per person? Name one alternative and its trade-off.
3. Say the two-sentence honest reading of the ancestry figure (distributions *and* accuracy).
4. Why report the PGI’s *incremental* R^2 rather than its R^2 alone? What would removing the PCs do to the coefficient?
5. What one variable changes to run everyone rather than 1-in-10? All 22 chromosomes?

34–40 minutes: save and finish

```
bash scripts/04_save.sh
```

In AoU mode the disclosure screen **blocks** the save on small-count findings; today’s tables are designed to pass, and the screen’s report will list your person-level files as workspace-only — read that list. Verify the cloud listing, **stop your cloud app** (VMs bill while idle), and finish the exit check: *I pulled real records, defended every cleaning step, counted every join, and put a real PGI in a model I can defend.*

Taking this to your own project

Change the concept ID, bounds, and one-row rule for your phenotype; swap in your trait’s published weights; set `SAMPLE_MOD=1`; loop `CHROM` over 1–22 and sum the `.sscore SUM` columns. Rehearse in the sandbox first — everything except the BigQuery step tests locally for free.